ENGR 852 Motion Estimator

V4 design, verification, and 14 nm implementation record | November 4, 2026

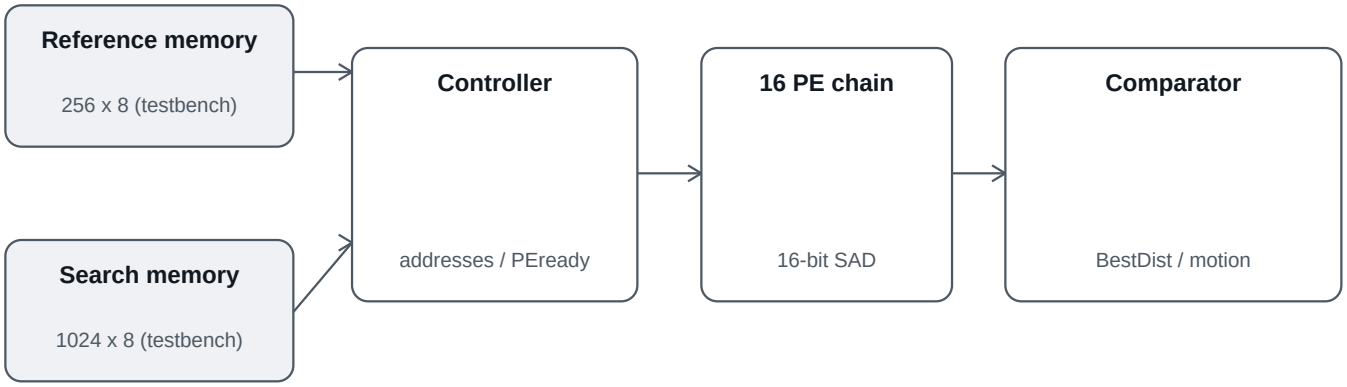
Document scope: actual VCS/Verdi verification, full-precision SAD correction, mapped SAED14 synthesis measurements, physical-flow checkpoint, and outstanding sign-off evidence. Measurements are labeled by their flow stage.

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1. Design target and memory boundary

The design compares an 8-bit grayscale **16 x 16** reference block against candidate positions in a **31 x 31** search window. A 16-PE systolic datapath performs absolute differences and accumulation while the controller generates search/reference addresses and candidate vectors. The comparator selects the minimum sum of absolute differences (SAD).

External testbench memories are NOT synthesized inside top.



Reference/search data and addresses cross the memory interface

Course setting	Required value
Clock frequency and cycle	520 MHz; 1.923 ns period
Input and output delay	0.500 ns each
Clock uncertainty	0.200 ns
Reference/search memory	Outside synthesizable top; testbench models drive R, S1, S2
SAD precision	16 bits to accommodate 256 x 255 = 65,280 (FF00)

The provided top-level RTL has memory-data inputs R/S1/S2 and address outputs AddressR/AddressS1/AddressS2; only the controller, processing elements and comparator belong in the physical chip.

2. Functional RTL validation

The original source ZIP was preserved separately. Baseline directed and randomized tests were executed with Synopsys VCS W-2024.09-1; Verdi W-2024.09-1 displayed genuine FSDB waveform data. The clock toggles every 0.9615 ns, producing the target period of 1.923 ns at timescale 1ns/100fs.

Stimulus / scenario	Observed result	Interpretation
Directed reference/search fixture	BestDist=00; X=7; Y=8; candidate count 271	Perfect match, signed vector (+7,-8)
Single-pixel +3 perturbation	BestDist=03; X=E; Y=D; candidate count 1542	Nonzero SAD, signed vector (-2,-3)
Upper-corner boundary	BestDist=00; X=7; Y=7; count 4111	Last candidate, signed vector (+7,+7)
PE / PEend / comparator directed	12 checks, 0 failures	Saturation, mux path, strict-less-than behavior
Randomized saturating-PE scoreboard	3 seeds, 24,576 PE comparisons, 0 mismatches	The original 8-bit *saturating* behavior matches its reference

Interpretation: the randomized PE scoreboard establishes agreement with the defined saturating 8-bit behavior; it does **not** prove that 8-bit SAD gives the correct minimum for every image. This motivated the separate full-precision correction.

2.1 Reproduced saturation defect

For the directed all-high-contrast fixture, all 256 reference pixels are 00 and every search pixel is FF. Each candidate has a true SAD of 256 x 255 = 65,280 (hex FF00). Original PEs saturate at 8-bit FF. The original comparator initializes BestDist to FF and updates only when newDist is **strictly smaller**, so no saturated candidate can be accepted.

Output	Original RTL / VCS observation	Corrected variant / VCS observation
BestDist	FF	FF00
motionX / motionY	X / X (unknown)	8 / 8 = signed (-8,-8)
completed	1	1
Comparator newBest	0; no valid winner	1 on first equal-SAD candidate

The 16-bit fix widens the PE accumulators, comparator datapath and best-distance register. The independent V3B hold variant also avoids output changes after completion; its directed run retained the corrected result for 13 further clock cycles. The professor source and teammate implementation were not overwritten.

3. Native waveform evidence

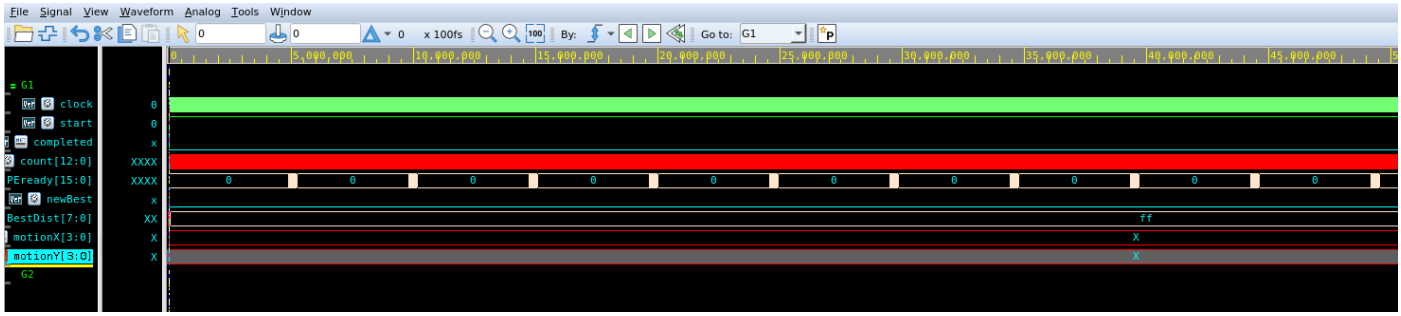


Figure 1. Original 8-bit VCS trace displayed in Verdi nWave. PEready pulses occur but newBest stays low; BestDist remains FF and motionX/Y are unknown.

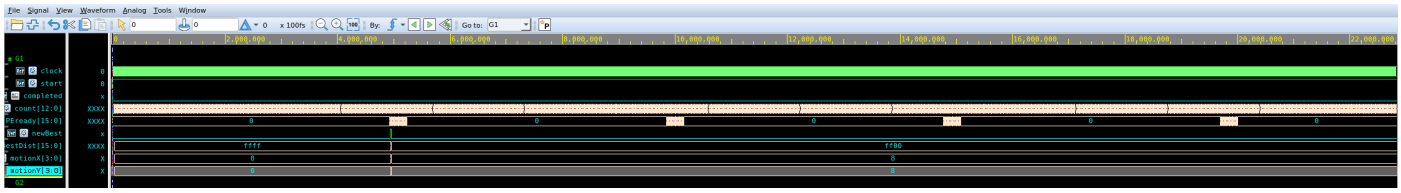


Figure 2. Corrected 16-bit Verdi nWave trace. NewBest updates BestDist to FF00, and the first candidate is encoded as 8,8 (signed -8,-8).

Both images were captured from the nWave application window without including unrelated desktop applications. Cropping was limited to the waveform panels; the original raw evidence is preserved in earlier V3 records.

As in the team controller slide, the signal list includes clock, start, completed, count[12:0], PEready[15:0], newBest, BestDist and motionX/Y. This contribution focuses on the comparator and SAD-precision failure rather than repeating the controller-row demonstration.

4. SAED14 Fusion Compiler synthesis results

The instructor-prescribed local Fusion Compiler setup loaded X-2025.06-SP2a and SAED14 RVT libraries. The example environment and motion-estimator mapped compile completed. The numbers below were read from native Synopsys reports and are **mapped-cell measurements**, not final routed core area.

Report metric	Original 8-bit RTL	Corrected 16-bit V3B
Mapped standard-cell area	881.96 um2	1,193.03 um2
Mapped cell count	1,693	1,905
Combinational cells	1,413	1,489
Sequential cells	280	416
Memory macros / black boxes	0	0
Compile-stage setup slack	First SDC run invalid input-delay collection	+0.06 ns (MET)

Corrected implementation area grows by **311.07 um2 (35.3%)** relative to the original. The 16-bit datapath preserves SAD values up to 65,280 without loss. Timing impact requires further optimization and post-route analysis.

4.1 Timing methodology

• Clock 1.923 ns and uncertainty 0.200 ns are set in the design constraints.

• External input delay and output delay are each 0.500 ns.

• An unsupported expression in the first motion-estimator input-delay command was corrected, and the reloaded timing path confirmed 0.500 ns input delay and 0.200 ns uncertainty.

• The measured +0.06 ns path is a compile-stage max-delay setup report. It is **not** a post-layout PrimeTime setup or hold measurement.

4.2 Power assumptions

The mapped flow can produce preliminary power reports, but a verified numeric final power figure and activity assumptions were not recovered in the evidence available for this report. A legitimate final power value must identify the cell library corner, clock rate, toggle/activity basis, and post-route extracted interconnect model. A nominal estimate is not inserted as a measured result.

5. V5 Physical Implementation Results

This section replaces the earlier V4 physical-status page. It describes work actually performed with Fusion Compiler and IC Validator on November 5, 2026. The original instructor RTL and team slides remain unchanged. The separate V5 experiment uses the corrected **16-bit SAD** implementation.

Step	Observed evidence	Status
Technology mapping	SAED14 RVT, corrected RTL, zero memory macro cells	PASS
Floorplan	35% target; 35.44% reported utilization	PASS
Placement	Successful place_opt_v5 design block and native tool log	PASS
Clock-tree synthesis	clock_opt_v5 saved; earlier 112.88% over-utilization resolved	PASS
Detailed routing	route_opt_v5 saved; zero route shorts and total zero routing violations	PASS
Final ICV DRC	242 rules checked; 18 violations across 3 rules	FAIL
GDS export	GDS generated; GDS-110 unplaced-cell warnings	REVIEW
SPEF extraction	Min, nominal and max routed SPEFs exported (about 8.8 MB each)	PASS
PrimeTime setup/hold	Full post-route reports generated for FAST/TYPICAL/SLOW	VIOLATIONS

5.1 Post-route area and dimensions

Quantity	Measured value	Measurement source
Routed cells	6,310 (5,861 combinational; 449 sequential)	Fusion route_opt area report
Standard-cell area	2,781.30 square micrometers	Fusion route_opt area report
Buffers and inverters	4,547 cells	Fusion route_opt area report
Memory macros	0	Fusion mapped netlist; off-chip/testbench memories
DEF die bounding box	92.510 x 50.800 micrometers	Exported floorplan DEF
Die bounding area	4,699.508 square micrometers	Calculated from DEF coordinates

Layout interpretation: standard-cell area measures the mapped cell footprints and differs from the enclosing die/core geometry. In particular, the exported GDS is not an accepted final clean layout while unplaced-cell warnings remain.

5.2 Power result: actual tool estimate

At the typical SAED14 corner (**0.80 V, 25 C**), Fusion Compiler reported total dynamic power of **0.728 mW**, leakage power of **0.000775 mW**, and total estimated power of **0.729 mW**. Activity was estimated through the tool's switching-activity propagation, not measured on physical silicon.

6. Extracted PrimeTime Timing and DRC

The required **1.923 ns period, 0.500 ns input delay, 0.500 ns output delay, and 0.200 ns clock uncertainty** were preserved in all runs. Fusion Compiler exported SDC version 2.2; a copy was adjusted only for PrimeTime SDC 2.1 syntax and the matching FF, TT or SS library. The original design constraints were not relaxed.

Paired PVT / SPEF	Setup WNS	Setup violations	Hold WNS	Hold TNS	Hold violations
FAST: FF 0.88V 125C / minTLU_125	+0.52 ns	0	-0.14 ns	-12.96 ns	402
TYPICAL: TT 0.80V 25C / nomTLU_25	+0.30 ns	0	-0.12 ns	-4.76 ns	120
SLOW: SS 0.72V -40C / maxTLU_-40	-0.06 ns	17	-0.09 ns	-1.24 ns	33

Interpretation: Fast and typical setup paths pass, but the slow setup group fails on 17 input-to-register paths. Hold paths fail at every corner. This is actual post-route, SPEF-annotated PrimeTime timing, not the earlier +0.06 ns compile-stage setup estimate.

6.1 Final ICV design-rule violations

ICV check	Violations	What fails
M2.S.2	6	Insufficient spacing for wide/parallel M2 shapes
M2.S.3	6	Insufficient extra spacing for another M2 width/run-length condition
M5.W.1	6	M5 widths outside the permitted discrete set
Total	**18**	**3 distinct rule checks; sign-off NOT passed**

The routed design's zero short / floating-route violations are not equivalent to an ICV DRC pass. ICV ran 242 checks and the final results file reports 18 violations. Unplaced cells identified by GDS-110 and incomplete filler-cell coverage must also be resolved before calling the layout complete.

6.2 Timing repair: native PrimeTime ECO

Using the course-provided **fix_eco_timing -type hold** procedure and the SAED14 ECO buffer set, PrimeTime generated **731 insert_buffer commands**, additional cell-sizing changes, and an estimated buffer-area increase of **412.70 square micrometers**. PrimeTime reported no remaining FAST hold violations in the *unrouted in-memory ECO model*. The changes were applied in a new Fusion Compiler block, **signoff_v5_holdfix1**, and routing was started. Final improvement requires completion of that route, new SPEF, final ICV DRC, and fresh three-corner PrimeTime max/min reports.

7. Requirements and Sign-off Decision

The course grading rubric specifies all major design elements correctly verified and implemented, completed Verilog testbenches, physical design, power/area, routed layout, DRC, parasitic extraction, and post-layout PrimeTime setup and hold. The following evidence-based audit distinguishes completed deliverables from acceptance gates that still must pass.

Submission item	Evidence available now	Rubric status
Debugged RTL and Verilog testbench	Separate 16-bit SAD RTL; VCS original fixture and 3/3 regression; saturation and completion-hold corner tests	VERIFIED for documented cases
Simulated waveforms	Genuine Synopsys Verdi FSDB captures plus local VCD and native VCS logs	AVAILABLE
Gate-level netlist and constraints	SAED14 full routed netlist and 520 MHz SDC; correct 0.5 ns delays / 0.2 ns uncertainty	AVAILABLE
Layout files, area, power	Exported GDS/DEF; 2,781.30 square micrometer cell area; 0.729 mW typical estimated power	NEEDS clean final GDS
Design-rule checking	ICV report contains 18 M2/M5 violations	NOT PASSED
Parasitic extraction	Routed SPEF at fast, typical and slow scenarios	VERIFIED, pre-ECO
PrimeTime setup and hold	Three-corner valid report set with quantified worst paths and violations	NOT CLOSED
Presentation and electronic report	V4 team and three personal slides; updated physical figures in this V5 report	READY for revision
Demonstration preparation	Verilog stimulus, VCS logs, Verdi captures, physical-flow outputs	SUPPORTED

7.1 Required final acceptance checks

• Finish the new ECO route and legalize all inserted cells, with no broken nets, shorts or floating routes.

• Rerun final ICV DRC until the applicable rule violations are zero; resolve GDS-110 unplaced-cell and filler warnings.

• Export the new gate-level netlist, layout and **new** extracted min/nom/max SPEFs after any ECO.

• Rerun PrimeTime max (setup) and min (hold) with actual extracted RC at all three corners; achieve nonnegative worst slack at each.

• Refresh final area and estimated power using the finalized post-ECO netlist and documented switching assumptions.

• Add final native layout and timing figures to the presentation and retain the complete scripts and source in the project package.

Conclusion: Functional correctness, complete routing and parasitic extraction are demonstrated. The current physical design is not yet a clean sign-off implementation, so no guaranteed grading percentage is claimed. The engineering record identifies the specific outstanding errors and preserves every successful and failed checkpoint.

Versioned evidence: github.com/huynguyen1017/motion-estimator-asic-lab-notebook-2026/docs/UPDATE-019_V5_POST_ROUTE_EXTRACTION_PT_ECO.md

V6 Addendum — Verification and GDS Audit

ENGR 852 Motion Estimator | November 6, 2026 | 520 MHz

This dated addendum supersedes earlier ambiguous BestDist captions and updates the GDS-placement checkpoint. Existing V5 timing and physical-design measurements remain historical values until a corrected, signed-off layout is produced.

A. BestDist depends on the stimulus

Stimulus	Expected BestDist	Meaning
Team perfect match	0000	Reference equals one search candidate; a 16-bit accumulator still returns zero.
Perturbed match	0003	The minimum candidate difference is three.
All-00 reference vs all-FF search	FF00	A separate worst-case directed test: 256 x 255 = 65,280.

The original eight-bit datapath saturates at FF for the extreme input. Its strict-less-than comparator has no valid new best against a reset limit of FF. The corrected sixteen-bit datapath selects FF00, with four-bit vector (8,8), signed (-8,-8), **only for that extreme stimulus**. No RTL change is needed to make the normal test return 0000; it already does so.

B. Source compatibility and verified regression

The current source preserves the original top/controller/PE/PEend/PEtotal/Comparator hierarchy, 16-element search pipeline, two-port search memory interface, candidate scanning, and signed vector encoding. The changes widen sum and comparator buses to 16 bits and qualify the completed outputs. Reference and search memories remain in the testbench.

Verification item	Observed	Outcome
Source match	Clean V6 top.v equals the reviewed verified RTL, byte for byte	PASS
Seven revision-aware simulation processes	7/7 exit successfully	PASS
Original baseline, three independent expected-value assertions	0000 / 0003 / 0000, corresponding vector checks	PASS
Corrected top-level, three directed checks	0000 / 0003 / 0000 with expected vectors	PASS
Intentional 8-bit saturation negative control	Original no-selection failure reproduced	PASS as negative control

V6 Addendum — Physical Sign-Off

C. GDS re-export from the existing V5 routed checkpoint

A clean native Fusion Compiler diagnostic on top/signoff_v5 located controller count_reg[0] as placed and returned **zero unplaced flat cells**. Placement legality completed. A fresh write_gds produced an approximately 80 MB GDS file, exit code zero, with **zero GDS-110 warnings**. The previous GDS output remains archived; the fresh file is retained on the engineering server.

This corrects the earlier GDS-placement uncertainty for that checkpoint. It does not prove that the layout is ICV DRC-clean, nor that future hold/setup repairs preserve routing legality.

D. Outstanding acceptance gates

Stage	Current evidence	Required for 90–100 rubric target
ICV DRC	18 V5 baseline violations; isolated repairs still nonzero	Final design with zero applicable ICV violations
SLOW setup	-0.06 ns / 17 paths. Trial ECO -0.02 ns / 2 paths, in memory only	Physically implemented, routed and extracted setup with no failing paths
FAST hold	-0.14 ns / 402 failing paths	Nonnegative hold slack and no failing paths
TYPICAL hold	-0.12 ns / 120 failing paths	Nonnegative hold slack and no failing paths
SLOW hold	-0.09 ns / 33 failing paths	Nonnegative hold slack and no failing paths
Final extraction	V5 SPEF exists for three corners	Regenerate SPEF, gate-level netlist, power and area after final ECO

E. Reproducibility and protection rules

Each repair must run from a preserved copy of the design block. Check executable paths, Tcl variables, library inputs, tool licensing, saved database blocks and exact SDC before execution. Inspect log return codes *and* violation/path counts. If DRC or timing becomes worse, reject the candidate and keep the last good checkpoint; do not advance based on process exit code alone.

The corrected V6 contribution presentation distinguishes normal 0000 from the separate FF00 stress case. Its full 19-slide copy preserves team slides 1–16 and scales the physical-results slide to the same canvas. No new timing or DRC success is claimed in those slides.

The project is not yet fully signed off. Completion requires actual zero-violation ICV and three-corner post-layout PrimeTime setup/hold after all changes, supported by matching final routed files and reports.

Verification details: docs/UPDATE-021_V6_GDS_REGRESSION_AND_SLIDE_CORRECTION.md